Project Handoff Document: Enterprise Conversational Workflow Platform for Nigerian SMEs

1. Executive Summary & Project Scope

This document outlines the architecture, feature set, and scope for a B2B SaaS conversational AI platform built specifically for the Nigerian market. The platform upgrades standard WhatsApp checkout bots into a full-scale automated CRM and reservation engine.

It is designed for two primary customer profiles:

- **Tier 1 (Mid-Market):** Logistics firms, HMOs, real estate agencies, and civic tech platforms that require data security and resilient human handoff protocols.
- **Tier 2 (Micro/Small Businesses):** Instagram vendors and retail merchants requiring automated checkouts without the risk of foreign exchange volatility.

2. Solution Offering & Value Proposition

Current global WhatsApp CRM solutions (like Wati.io) fail to account for local infrastructural and economic realities. This platform introduces hyper-localized solutions to these core problems:

- **Infrastructure Resiliency:** Standard synchronous webhooks drop when users switch apps to complete bank transfers. This platform introduces an asynchronous "anti-ghosting" queue to prevent lost transactions.
- **Hyper-Localized NLP:** Standard LLMs fail to parse Nigerian Pidgin and commercial slang. This system uses a custom intent router to understand local dialects natively.
- **OPEX Protection:** Global SaaS providers bill in USD, causing pricing volatility for local merchants. This platform integrates with African Business Solution Providers (BSPs) to offer local Naira billing and hard-capped LLM token controls.

3. Core Deliverables (Features & CRM Engine)

A. Dynamic Lead Qualification Engine

- **Intent Scoring Matrix:** The NLU evaluates real-time scores based on the customer's urgency, budget, and technical requirements.
- **State Machine Routing:** Hot leads are pushed to instant checkout workflows, while complex technical queries trigger deep document retrieval via a Retrieval-Augmented Generation (RAG) pipeline.
- **Automated Nurturing:** Cold leads receive dynamically generated value propositions.

B. Reservation & Appointment Management

- **Atomic Slot Booking:** The database uses strict constraints to mathematically eliminate double-booking race conditions.

- **Anti-No-Show Queue:** Calendar slots are temporarily held in an asynchronous queue for 10 minutes and automatically released if the payment deposit webhook is not received.

C. Asynchronous Checkout ("Anti-Ghosting")

- **Decoupled Webhooks:** The platform immediately sends a 200 OK response to Meta to prevent timeout errors.
- **State Persistence:** Active chat states are preserved even if the customer leaves the app to make a transfer, proactively pushing the receipt into the chat once cleared.

D. Resilient "No-Repeat" Human Handoff

- **Offline Detection:** The system monitors local agents via heartbeats; if an agent drops offline due to local power or broadband failures, the bot queues the ticket asynchronously.
- **Context Injection:** An AI-generated summary of the customer's intent is injected into the inbox to prevent agents from asking repetitive questions when they reconnect.

4. Technical Architecture & Stack

The platform is designed to handle high concurrency and event-driven workflows while maintaining ledger accuracy.

- **Core Backend Structure:** A modular monolith deployed on high-performance event-driven runtimes (such as FastAPI with ASGI, or Swoole/RoadRunner).
- **Queue & Persistence:** In-memory stores (Redis) handle fast asynchronous queuing and state management.
- **Database & Ledger:** PostgreSQL serves as the relational database. It utilizes `TSTZRANGE` data types and `GiST` exclusion constraints for the calendar holds.
- **Ledger Validation:** Deterministic data validation (e.g., Pydantic in Python) enforces that $Assets = Liabilities + Equity$ before database commits. An AI fallback explains payment discrepancies to merchants if a webhook payload is mismatched.
- **AI/NLU Stack:** Python-native tools (spaCy or Rasa) power the deterministic fast-path intent router, while frameworks like LangChain or LlamaIndex manage the enterprise LLM escalation.

- **Merchant Interface:** A cross-platform mobile app built with Expo and React Native enables staff to manage queues, view receipts, and update local inventory on the go.

5. Compliance & Security Requirements

- **NDPA Compliance:** Explicit opt-in conversational flows are required for marketing broadcasts, backed by immutable audit logs.
- **PII Sanitization:** Pre-processing middleware must automatically detect and redact sensitive local data—such as Bank Verification Numbers (BVNs), account numbers, and passwords—before any prompt is routed to third-party LLMs.